

THE O. S. U. Naturalist

PUBLISHED BY
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Volume 1.

December, 1900

Number 2

COLUMBUS, OHIO

PRESS OF HANN & ADAIR

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Title: The O. S. U. Naturalist, Vol. 1, No. 2, December, 1900

Creator: Ohio State University. Biological Club

Release date: April 6, 2023 [eBook #70477]

Language: English

Original publication: United States: The Biological Club of the Ohio State University, 1900

Other information and formats: www.gutenberg.org/ebooks/70477

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*** START OF THE PROJECT GUTENBERG EBOOK THE O. S. U.
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*THE O. S. U.
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A journal devoted more especially to the natural history of Ohio. The official organ of THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY. Published monthly during the academic year, from November to June (8 numbers). Price 50 cents per year, payable in advance. To foreign countries, 75 cents. Single copies 10 cents.

Editor.

JOHN H. SCHAFFNER,

Agent.

F. J. TYLER, *Subscriptions.*

R. F. GRIGGS, *Advertising*

Address

THE O. S. U. NATURALIST, Ohio State University,
COLUMBUS, OHIO.

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Entered at the Post Office at Columbus, Ohio, as second-class matter, November 20, 1900.

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DECEMBER, 1900.

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FEEDING HABITS OF THE SCARLET FLAMINGO.

J. C. HAMBLETON.

One of the favorite winter haunts of *Phoenicopterus ignipallatus* is found on the sandy beaches that abound on the west coast of South America, at about latitude 42° South.

On the north coast of the large island of Chiloe, there are several places of this sort where these beautiful birds may be seen in flocks of hundreds during the months of June, July and August, the winter months in that region.

I remember the first time I walked over one of their favorite resorts. It was on the Pudeto river, near the small town of Ancud. The tides here are rather high owing to the formation of the bay, and as a consequence it enters the river and floods great stretches of sand that border the left bank. As the tide goes out the flamingos may be seen here by hundreds. The first time I visited the place the tide had been out some hours and there were no birds to be seen. I was disappointed for the trip had been made for that special purpose.

However my attention was soon attracted to long rows of small hillocks of sand, or rather, to be more exact, circular ditches in the sand that appeared to have been made while the water was still present. These were about two or two and one-half feet in diameter by five or six inches wide and three or four inches deep. This, of course, gave the central portion the appearance of a small hill about eighteen or twenty inches in diameter. Upon inquiry I could get no information—no one had any idea how or by whom they had been made.

A few days later the mystery was solved when a second visit was made to the place at a more propitious moment.

Upon approaching to within a few hundred yards of where a regiment of these scarlet beauties was lined up, the birds took flight and it was then that

I discovered who were the authors of my mysterious little hills.

By subsequent observations I found that the birds took their stand in the water when it was about a foot and a half deep, and at more or less regular intervals about eight or ten feet apart. Here they remained stationary and turned round and round with their heads under water, catching the small crustaceans that seem to be their principal diet. The form of their beak is such that when it is placed on the ground the upper mandible is underneath. This being large and strong, soon opens up the circular depression that first called my attention. Before the tide is all out they usually leave because the crustaceans have by this time hidden in the sand.

The flamingo frequents this coast only during the winter months and consequently does not nest here, nor is it known to nest west of the Andes mountains. Their nests and young, however, have been observed in great numbers in the small lakes of brackish water that abound on the plains of Patagonia east of the mountains. Undoubtedly these are the same birds that spend their winters in Chile, the lofty Andes proving no barrier to their flight. There are many roads by which they can pass, the mountains being intercepted by frequent rivers that empty into the Pacific, and have their origin beyond the snow-covered Andes, in the plains of the Argentine Republic.

AGGRESSIVE CHARACTER AND ECONOMIC ASPECT OF THE WHITE HEATH ASTER.

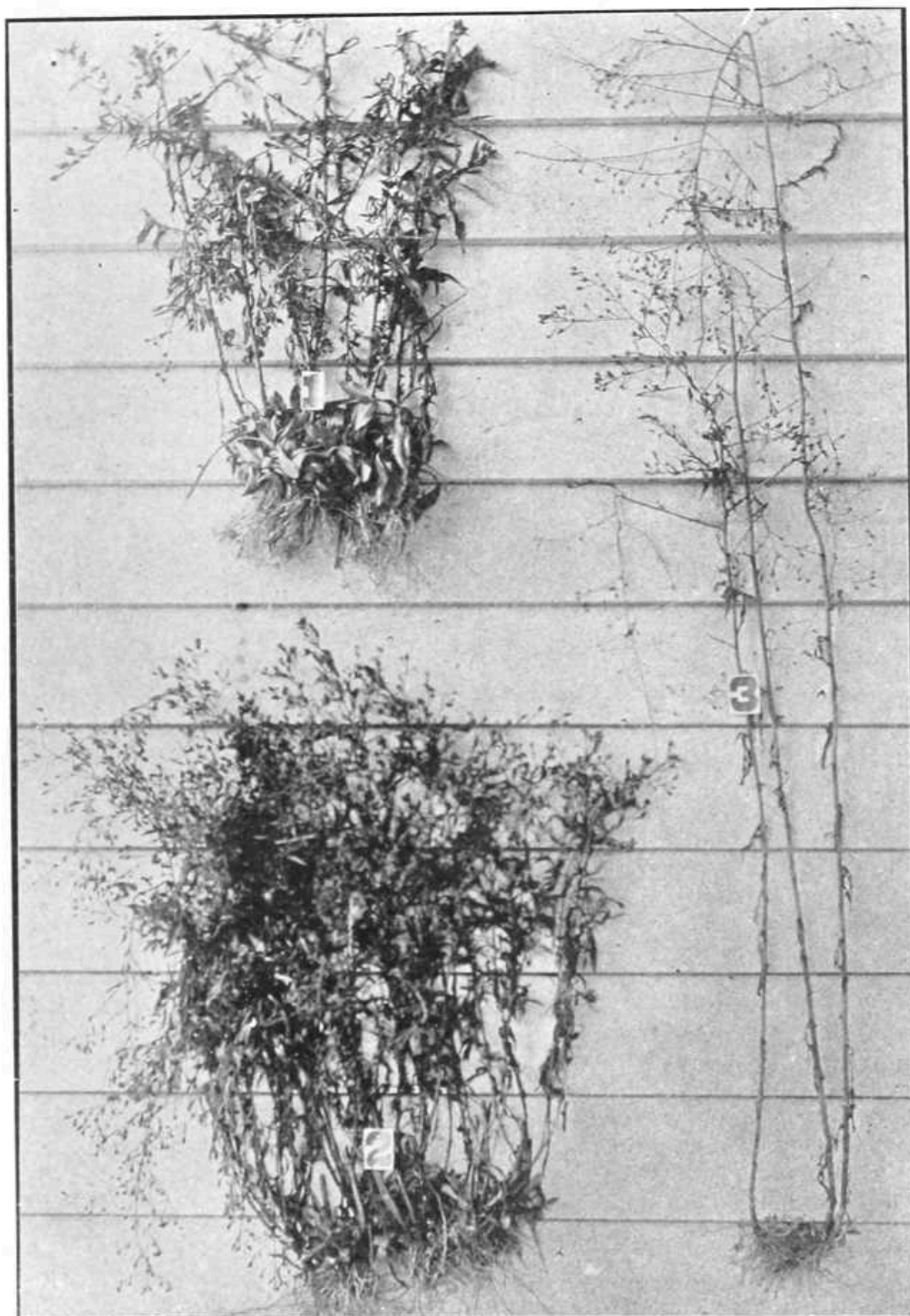
W. A. KELLERMAN.

(Plate [3.](#))

The White Heath Aster (*Aster ericoides*) is an indigenous species whose distribution is given in our manuals as “Canada, Florida, and the Mississippi,” “Maine and Ontario to Florida, west to Wisconsin and Kentucky,” and “South New England to Minnesota and southward,” the variety *pilosus* “mainly in the Western States.” It is one of the commonest Asters throughout Ohio, occurring doubtless in every county in our State. The variety *pilosus* seems to be the common form in our region, and may be seen growing in rich and poor soil with almost equal thrift, and occurring in all habitats except the woods and swamps.

Its capacity for adaptation to the advance of civilization is remarkable, and this occasions the remark now very generally heard among the farmers that it is a “new weed in the region,” “not known here five years ago,” “just came all at once,” “the latest and worst weed we have,” and other expressions of similar import. As a matter of fact, the roadsides in many places are lined with it, fields with a poor stand of clover, timothy, or blue grass are completely covered with it, and all waste places, vacant lots, and neglected spots are profusely decorated with the same.

Plate 3.



KELLERMAN ON ASTER.

The plant is a rather coarse weed; but in spite of this fact it is somewhat attractive because of the masses of green foliage and the white flowers that become prominent before the summer is gone, and last throughout the early and middle autumn. The stems are tough and wiry and this gives the local name "Steelweed," a common designation in Adams County and adjoining regions. It is said by some, however, that this name is given it "because the flowers are the color of bright steel." Another name frequently applied in the localities mentioned is "Bee-plant" for reasons suggested in the name itself, and still another is "Stickweed," for which I could learn no explanation. Other common names which Britton enumerates are Frost-weed, Michaelmas Daisy, Farewell Summer, White Rosemary, Dog-fennel, Mare's-tail, and Scrub-bush.

Though complaint against this plant is universal in some sections, it is not, I think, well founded in all cases. It has some merits now and then acknowledged by those who are close observers. The allegations pro and con may be summarized as follows:

First, the statement is made that it is "driving out every other grass" and "invading" the whole country. It is certainly more abundant than it was before the country was cleared and cultivated; yet after all but little of it is seen in good pastures and vigorous meadows, and none at all in ground that is under thorough and constant cultivation. It has not the aggressiveness possessed by some of our weeds, but it does quickly take possession of neglected and fallow ground. It does not spread extensively or rapidly by underground stems as do some of the Compositae. It has simply short rootstocks for this purpose. Its mode of multiplication by this means is illustrated in the figures shown in Plate 3. These are from photographs taken late in November, and indicate the preparation the plant makes for the next season's work. The specimens numbered 1 and 2 had been mowed to the ground during the summer. But this instead of killing the plants stimulated their propensity to vegetative multiplication. The result was therefore the opposite of what the farmer intended. Figure 3 shows a plant undisturbed during the growing period, and its energies active and latent were almost entirely exhausted in producing flowers and seed. Let the

plants alone then rather than shear their tops, and the sooner will they exhaust themselves.

It is true, as the figures plainly suggest, that this Aster is not a difficult one to eradicate. While the roots are numerous, they are not long; and even the shallowest plowing or ordinary cultivation will effectually destroy the plant. As to multiplication by seed germination, it needs simply to be remembered that good or even fair cultivation of the soil will prevent this weed from growing, and that many species of weeds will get in old meadows and pastures as rapidly as the cultivated grasses are killed by excessive grazing or the casualties of season and climate.

In the second place the weed is charged with the heinous crime of “killing stock.” Thorough inquiry in different localities established the fact that this plant, eaten to considerable extent late in the season by cattle and horses it is true, does damage perhaps only as the consumption of an excessive amount of almost any kind of dry and comparatively innutritious vegetable matter might do. It is said to be especially binding, and the constipation no doubt was a factor in bringing about the fatal results that were cited. While stock will eat the plant when at hand they take but little of it if nutritious grasses can be found. A very intelligent and observant farmer, however, was seen cutting and burning the plants which covered his pastures to save his stock—his neighbor by carelessness in this respect, he averred, having lost some valuable horses.

On the other hand this White Heath Aster is an important bee-plant. Bees will “work on it the whole day,” and the plant is in bloom from middle or late summer to late autumn. The honey made is white, and has a strong tendency “to turn to sugar.” One farmer who has two hundred and fifty stands of bees, now that this Bee-plant is well established as a sure crop, will sow no more buckwheat for his bees.

I have said this species is becoming excessively abundant in some (hilly) portions of southern Ohio. It can well be regarded as “a great boon” merely because it is a soil-binder of marked efficiency. It prevents the destructive washing of the hillsides in the Fall, open winter and early spring. Such a plant would not be needed to a great extent, were methods and habits of cultivation perfect or in a high state of development; but this phase of the economic aspect of the case must at present be insisted on.

Finally it may be said that as a fertilizer this Steel-weed takes a high rank. It is regarded by observant farmers as but slightly inferior to a crop of clover. It does not decompose when turned under as quickly as clover, but that it yields plant-food and answers well the mechanical purposes of a coarse fertilizer, testimony is unanimous and apparently conclusive.

EXPLANATION OF PLATE 3.—*Aster ericoides pilosus*, reproduced from photographs taken late in November. Figures 1 and 2 show plants with abundant, and Figure 3, with few young shoots close to the ground. Plants shown in Figures 1 and 2 had the tops removed in summer. Figure 3 shows the common appearance at the end of the growing season of undisturbed plants.

GEOPHILOUS PLANTS OF OHIO.

F. J. TYLER.

Geophilous—meaning earth loving—is a term which has recently been applied to such plants as have some special adaption, which enables them to withdraw beneath the surface of the ground when adverse conditions, such as extreme heat and drouth, cold, etc., overtake them. Such adaptations may be classified as

Rhizomes,
Bulbs,
Corms,
Crowns.

Rhizomes are underground stems and like other stems may be simple or branched. The branched rhizome is, however, the most common form since it combines vegetative reproduction with the other advantages of a rhizome habit. The Brake Fern (*Pteris aquilina* L.) is an example of a much branched rhizome and Solomon's Seal (*Polygonatum biflorum* Ell.) of a nearly simple rhizome.

The stem of a rhizomatous plant may remain permanently underground, as is the case with all ferns except the tree ferns of the tropics. An annual stem is, however, usually sent to the surface and this may be a lateral branch from the main subterranean stem or it may be a continuation of the rhizome, in which case the next year's rhizome will be a lateral branch and thus the whole rhizome will be made up of a number of distinct segments. Various members of the *Iris* group are good examples of this. In at least one genus of Ohio plants—*Smilax*—there are some members having both a perennial woody stem and a well developed rhizome. It may be that these plants are leaving the rhizome habit and are taking up the woody stem habit.

Perhaps in most cases rhizome plants became such through the gradual covering of trailing stems. It is a protection and a saving of building material to a plant if its stems are trailing or creeping, still more so if they

are covered by leaf mould or soil. If rhizome plants were once trailers there should be every gradation between the two and so we find. The Trailing Wahoo is a good example of this, since some of its stems are often covered by leaf mould or soil while others are on the surface or some inches above. Many of the Ericaceæ are in this transition stage between trailers and geophytes. The Wintergreen (*Gaultheria procumbens* L.) has a long, creeping stem which is often or usually covered by leaf mould. It roots freely and sends up perennial woody branches to the surface. It is hard to say in such cases whether the plant is geophilous or not.

Many geophilous plants of the rhizome type were doubtless once crown formers and here again we find a transition stage which contains every gradation between the two groups. The Compositæ are mostly crown formers, but some are true rhizome plants and some are transitional.

The advantages of a rhizome habit are very apparent. The first and most important advantage is the protection from frost which this habit affords. By taking up this habit many plants have been able to withstand a climate, which would otherwise prove fatal. The Alpine Willow is an example. Rhizomes are often storehouses for food and become swollen and distorted in consequence. Vegetative propagation is usually combined with the geophilous habit and with great advantage to the plant. In most cases a rhizome dies off at the back as fast as it grows in front so that any part of it lives a definite number of years. The individual segments of the Solomon's Seal, for instance, live from three to five years. In this way a branch soon becomes a separate plant. In some cases, however, the rhizome may live for many years and thus hundreds of seemingly independent plants may be connected beneath the surface of the ground. The Brake Fern (*Pteris aquilina* L.) is of this class and an entire hillside may be covered with a much branched specimen of this plant.

A plant which has no means of migration when it has exhausted the nearby food supply is manifestly at a disadvantage when compared with a progressive rhizome plant which moves every year into a new and fresh location. To be sure, the distance it travels may not be far but it is enough to remove the plant from an exhausted position and from its wornout and useless tissue. Thus this group of plants may be said to have found the secret of potential immortality, for, unless some catastrophe overtakes them, they may live indefinitely and remain young. It is interesting in this

connection, to note how far some of these plants travel in a century. This may be calculated in a general way by measuring the annual growth in length of the rhizome. Solomon's Seal travels from twelve to twenty feet in this length of time, *Uvularia perfoliata* L. from eight to ten feet, *Onoclea sensibilis* L. from three hundred to five hundred feet, and others still farther.

The Iris group are exceptions, in that they travel in a circle. The reason seems to be that the lateral branches which continue the rhizome from year to year mostly arise on the same side of the terminal bud, so that each branch goes off at a slight angle to the former branch. The degree of angle determines the size of the resulting circle. One class of rhizome plants is very distinct and requires especial mention. This class may be termed upright or retrogressive rhizome plants. The upright rhizome may originate from a progressive rhizome, or from a crown former or in some other way. *Trillium nivale* Riddell is in a transitional stage between the progressive and retrogressive classes, since the large rhizomes are upright and the young lateral branches are progressive until they have traveled some distance away from the parent rhizome, when they too, become upright. The lower Ferns (*Ophioglossaceæ*) belong to this class. The disadvantage of this habit is that the rhizome will soon grow out of the ground and be in a very exposed condition. To counteract this tendency the roots of these plants are usually strongly contractile and pull the rhizome down into the ground as fast as it grows out. Skunk Cabbage (*Spathyema foetida* (L.) Raf.) has an upright rhizome and root contraction is very marked. The very apparent disadvantages of the retrogressive or upright rhizome habit have made this class very few in number compared with the progressive rhizome class. In Ohio there are about 475 species of rhizome plants and less than twenty-five of these belong to the retrogressive class. This class is closely related to the corm plants, indeed, all that is needed to make the typical corm out of a retrogressive rhizome plant, such as *Trillium nivale*, is to shorten and make more definite the annual growth of the rhizome. The bulb is usually a very short, upright rhizome with many thickened scales. The bulb of *Lilium martagon* is of this kind but that of *Lilium canadense* is more closely related to the progressive rhizomes. The parent bulb sends out one or more thick rhizomes which grow outward if the bulb is at the normal depth, downward if the bulb is too near the surface of the ground and the new bulbs are formed by the shortening of the outer end and the growth and thickening of the scales of the rhizome.

Both bulbs and corms may be regarded as rhizomes modified to suit peculiar conditions, such as a long, dry, heated period alternating with a short, rainy period. A plant to survive under such conditions must be able to start up very quickly as soon as the rains come, and flower and mature its seeds before the drouth again overtakes it. A large amount of food material must be stored up by the plant in order to do this, and the food material must be kept from drying or burning up during the heated period. Bulbs and corms, protected as they usually are by dry and coriaceous coverings, answer these requirements and are usually abundant in localities where these conditions obtain. Bulb and corm plants are also well fitted to live in dense woods where the light is soon shut off in the Spring by the expanding leaves of the trees. They are able to spring up very early, flower and ripen seeds before the light is shut off. The food supply which enables them to do this is often protected by acrid or poisonous principals developed in the bulb or corm. Pepper-root (*Dentaria laciniata* Muhl.) and Jack-in-the-Pulpit (*Arisaema triphyllum* [L.] Torr.) are examples.

Crown plants, while not true geophytes, are often closely related to rhizome plants and may be regarded as transitional. They are formed by the freezing back of the upright stem to the surface of the ground, and the survival of the short stem beneath the surface until the next Spring when it sends out branches from adventitious buds. In this way several branches are sent up where there was one before, and, as this crowds and injures the plant, these branches usually move out some distance from the base of the parent plant before coming to the surface. The connection with the main stem is often severed, and thus many new plants are formed. All this rarely takes place in the Spring but has been shifted back to late Summer or Fall by the parent plant. Often a food supply is stored up for the young plants by the parent. *Helianthus tuberosus* L. is a good example.

Vegetative propagation is brought to its highest development in this class and they become our worst weeds.

NOTES ECONOMIC AND TAXONOMIC ON THE SAW BRIER, SMILAX GLAUCA.

W. A. KELLERMAN.

(Plate [4.](#))

In a recent trip through some of the southern counties of the State my attention was arrested by the enormous quantity of *Smilax glauca*—Glaucous-leaf Brier as given by Britton in the Illustrated Flora—but generally and appropriately called in these regions where so abundant, the Saw Brier. In the sandy soil of Hocking County, thence southward to the Ohio River this plant may be seen growing in field and pasture, by roadside and on hillside, and everywhere except in wet soils and dense woods. It climbs over fences and high bushes, displaying its bright foliage of lively green, more effective by contrast with the abundant white bloom on the under side. In the Autumn it presents showy wreaths of black but glaucous-coated berries and the most gorgeous coloration of foliage. The leaves remain for the most part late in Fall and Winter, and for brilliant and delicate shades of rose and red are not surpassed by any plant of our entire flora. The forbidding aspect of the long, wiry stems, with their bristly covering of long, saw-like or needle-shaped prickles, serves also to distinguish this plant even among the attractive associates of its kingdom.

A BAD WEED.—As a weed this species here stands at the head of the list. Its horrid prickles make it one of the most disagreeable plants with which to come in contact. It revels in the pastures and clambers over the fences; it flourishes in the meadows and fields, and no ordinary practice of crop-cultivation interferes with its luxuriance. One can readily see that it is not carelessness on the part of the farmer that suffers half or still larger portions of his fields to be covered with this pestiferous vine. No other weed is seen in the area and therefore he has been diligent and careful in his tillage. The meadows even if twice or thrice mowed in a season will yet contain year to year the same quantity of Saw Brier. The stems spring up quickly, and grow “a foot in a night” the people say; surely the Saw Brier is the freshest plant in the field. In a case specially noticed a garden spot had been put in

cultivation in 1873, and has been continually and thoroughly cultivated every year since, yet the Saw Brier is there to-day.

THE UNDERGROUND PARTS.—This tenacity of life and luxuriance of growth can be understood when the underground parts are examined. There are numerous irregular and often large tubers or enlargements which serve as the capacious storehouse of nourishment. They are often of fantastic shape. Various forms are shown in figure 1, plate 4. These occur at irregular intervals on the long and tortuous subterranean stems. It is said that they may be found several feet below the surface, though the eight specimens shown on the plate were found at a depth of six to twelve inches. If they all could be removed from the soil the weed would of course be practically annihilated. But when found at a depth of several feet—as seen sometimes in making excavations for foundations, walls, etc.—it is evident that the farmer will have to make extraordinary and long-continued efforts to destroy this pest. The less courageous may well be appalled in contemplating the herculean task. Fortunately swine are fond of the nutritious tubers, and voraciously devour them when they are given the freedom of the field and allowed to indulge in their natural propensities. Heavy coating of manure and winter plowing are also indicated.

VARIATIONS IN THE LEAVES.—This form is easily recognized among the several species of *Smilax* indigenous to Ohio, though the leaves vary in size and shape to a remarkable degree. A large number of the common forms are shown in figure 2, plate 4. They are sometimes very broadly cordate-oval, wider than long; often ovate-cordate, oval or ovate, lance-oval, oblong to oblong-ovate, broadly to narrowly lanceolate and even linear; they are mostly cuspidate at the apex, in some cases tapering to acute or sub-acuminate. The base is mostly cordate and subcordate, but occasionally tapering. It is seldom that forms approaching halberd-shape occur in our region. An inspection of plate 4 will illustrate these several forms. The twig with fruit marked 6, bears a very common form of the leaf as does also the one marked 1. The broad-leaf form is not uncommon, but the very broad-leaf as seen in specimens marked 3 and 5 is of much less frequent occurrence. The very narrow leaves are as a rule borne on short stems—such as have developed in fields and meadows where the earlier stems have been destroyed or disturbed. But often on such stems (marked 1) the leaves are broad and have the natural shape.

SIZE OF LEAVES.—The size of the leaves is strikingly variable. The specimens shown on plate 4 being reproduced from photographs exhibit correctly this variability. Careful measurements also were made of leaves taken at random from hundreds selected to illustrate this point. These since they give both the length and width of the leaf demonstrate the shape as well as the size. The measurements are in decimeters and one hundred of them are as follows:

12×11.5	12.5×9.5	8.5×6	9×2	9.5×3.8	7.8×2.2
11×10.5	12×9.5	7.5×5.5	9×3.5	8×4	7.3×2.8
12×11	12×11	8.6×5.4	8×1.5	7×2.8	6×1.7
13×10.6	11×11.8	8×6	6×0.7	8×3.8	5.3×1.6
11×12	12.2×9.8	9×6	8×1.5	4.5×1	8×3.4
12.5×11.2	11.5×10	8.5×6	7×1.8	7×2.2	7.5×2.5
12.2×10	13.5×10.5	8.5×5.8	6.8×2	7×3.2	6×2.5
12×11	11.5×9.6	9.8×6.4	7.5×2.2	6.2×2.5	6×2.8
11×11	8×5.5	9×7.3	5.5×2.5	8.2×2.8	6×2.6
11.5×11.1	9×6	8.3×5.8	7.6×2.2	6.5×2.2	7×2.8
11×10.8	9.2×6.5	9×6.5	7.8×2	5.5×2.9	8.4×3
12×11.5	8×5.5	9.9×7.6	6.5×2	7.3×2.7	9.8×3.5
12×10.6	9×7	8.2×4.6	7×0.8	8×1.8	10×3.5
13×11	8.5×7.5	8.5×6	8.4×0.7	9.5×1.9	9.6×3.8
12×11	7.5×4.5	8.8×6	8×1.4	9.5×2.2	7.5×1.7
11.5×10.8	8.5×6.2	9.5×7	8.5×2	8.4×2.2	9.4×2.5
12×11.3	9×7	11.1×2.2	9×3.5	8×2.5	

DESCRIPTION OF THE LEAF.—Neither the description as given originally (1787) by Walter nor those contained in our Manuals give any intimation of such variability as actually occurs. Some of them are as follows: Walter says “foliis oblongo-cordatis”; Wood, “ovate, finally nearly orbicular, abruptly contracted at one end”; Gray, “ovate, rarely subcordate, abruptly mucronate”; Britton, “ovate, acute or cuspidate at the apex, sometimes cordate at the base.” I would suggest the following as applicable to the Ohio specimens: *Leaves mostly ovate, often broadly oval (occasionally broader than long), sometimes oblong-ovate, varying to lanceolate or even linear; the base mostly subcordate but often cordate or even cuneate; the apex cuspidate to acute or sub-acuminate.*

SMILAX SPINULOSA.—Britton and Brown in the illustrated Flora, 1:440, appends to the description of *Smilax glauca* the following paragraph: “*Smilax spinulosa* J. E. Smith, is a form with numerous small prickles on the lower part of the stem, and more elongated, sometimes halberd-shaped leaves. It occurs in southern New York, but is not well understood.” However *Smilax spinulosa* is given by these authors as a synonym of *Smilax glauca*, which according to my judgment is correct.

Plate 4.

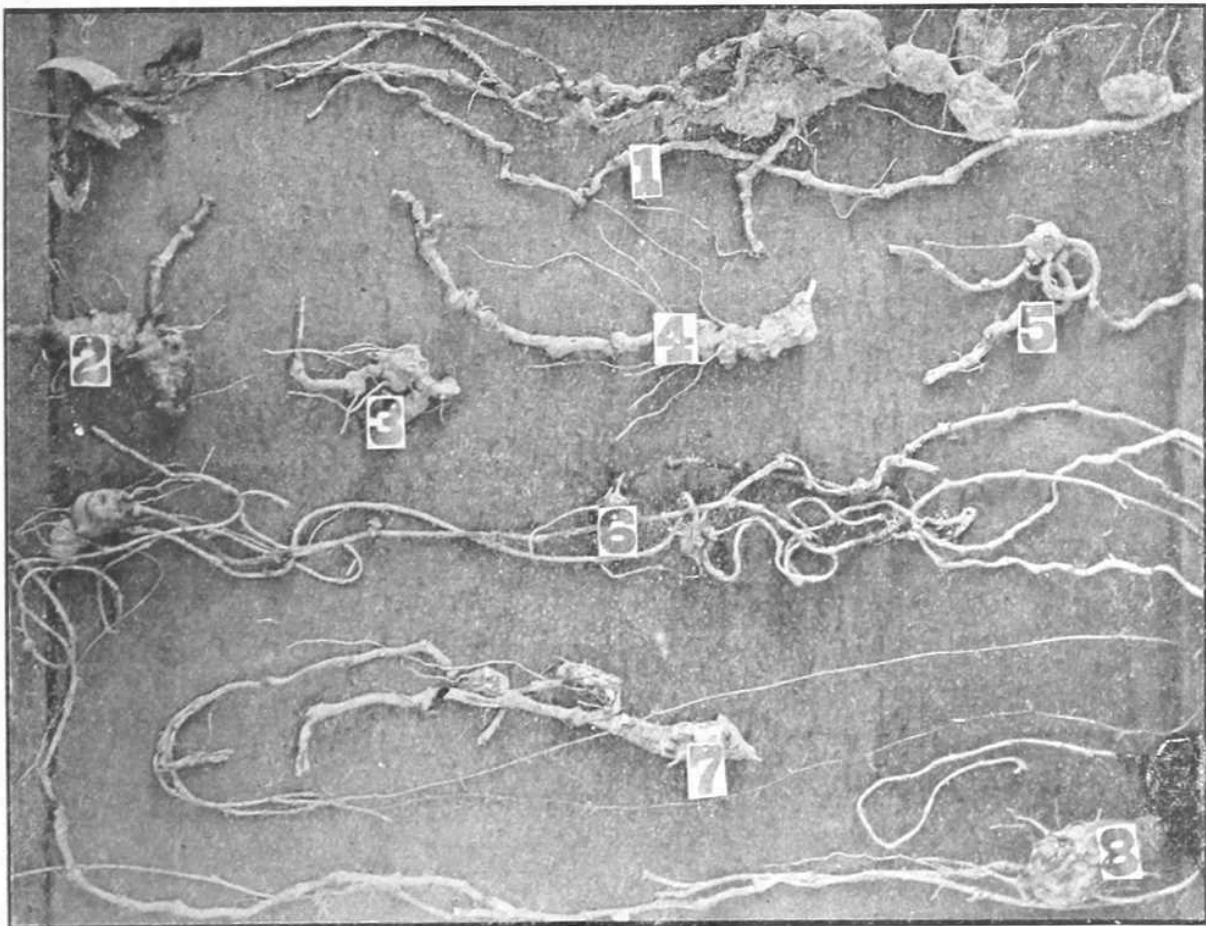


Fig. 1.



Fig. 2.

KELLERMAN ON SMILAX.

SMILAX SPINULOSA AN EXTREME FORM OF S. GLAUCA.—The material secured recently fortunately clears up the case unless I misapprehend the purport of the above quotation. The “more elongated leaves” spoken of may find their counterpart in the figures on plate 4, and yet there can be no question that they belong to *Smilax glauca*. The twigs bearing them were in many cases found *attached to the same underground stems* that bore the broad leaves. Even at a glance the identity of the specimens in the field could not be mistaken; all their characters showed that they were really *Smilax glauca*. In herbarium specimens that have been preserved every gradation may be seen between the extremes shown in the plate. These specimens also show in some cases underground stems that bear both leaves and twigs with leaves of the broader form. As a rule in the cornfields where

the soil is not rich and crop-cultivation has been diligent the narrower leaves (on shorter stems) are common. In richer cornfields, and in meadows, especially if quite fertile, the short stems are clothed with the broader leaves. Abundant evidence was at hand to demonstrate that this form with "more elongated leaves" (*S. spinulosa*) is directly connected with the form called *S. glauca* by the taxonomists. Its peculiarities are doubtless referable to the special environment; in no case could these apparently aberrant specimens be called a specific or even varietal form.

EXPLANATION OF PLATE 4.—*Smilax glauca*. Figure 1: Eight specimens of tubers reproduced from photographs and much reduced. Fig. 2. Twigs (1-6) bearing leaves of varying shapes also separate leaves (7-56) illustrating variation in shape and size.

MEETINGS OF THE BIOLOGICAL CLUB.

OCTOBER 1ST, 1900, BOTANICAL HALL.

This evening was spent in the giving of reports on work done during the summer vacation. Osborn, Mills, Griggs, Tyler, Miss Dufour, Schaffner and Hine gave reports. Seven names were proposed for membership.

The motion by Professor Osborn to hold the meetings of the current year in Biological Hall was carried.

Hine, Morse and Miss Dufour were appointed a nominating committee to select officers for the year.

NOVEMBER 5TH, 1900.

Members assembled in Biological Hall. President Schaffner called the meeting to order. Officers were elected as follows: Osborn president, Tyler vice-president, Hine secretary.

Long, Wyman, Ball, Hambleton and Dunlap were elected to membership.

The retiring president, Professor Schaffner, gave the address of the evening. He had for his subject "The Life History and Cytology of Erythronium."

Landacre and others took part in a discussion of the paper.

J. H. S.

NEWS AND NOTES.

The tenth annual meeting of the Ohio State Academy of Science will be held at the Ohio State University, Columbus, on December 26 and 27, 1900.

PETIOLES OF THE COTTONWOOD.—The leaves of the Cottonwood as well as some other species of *Populus* have an interesting light relation. If one looks at a vigorous shoot, the leaves are seen to be arranged in the profile position around the stem. This is accomplished by means of a very simple device. The petiole is much flattened next to the blade and is quite flexible. The flattening is transverse to the plane of the blade and on this account the leaf will assume the vertical position, no difference in what way it is attached or twisted. This adaptation is also responsible for the musical rustle one hears when resting in the shade of the Cottonwood. The leaves are very smooth and nearly alike on both sides. Because of these and other adaptations, the Cottonwood is one of the most successful of our semi-xerophytic trees and one of the last stragglers to be found in the western part of the great plains.

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Transcriber's Notes

- pg 15 Changed: Assciate Editors
to: Associate Editors
- pg 18 Changed: catching the small crustacians
to: catching the small crustaceans
- pg 28 Changed: J. S. H.
to: J. H. S.
- pg 29 Changed: Entomological Suppies
to: Entomological Supplies

*** END OF THE PROJECT GUTENBERG EBOOK THE O. S. U.
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